

Henry Lee

Backend Team Lead · Distributed Systems · 8 years in software

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8 yrs IN SOFTWARE · 6+ AT SCALE	50k → 400k+ DAILY ACTIVE USERS	Millions / day TRANSACTIONS	100+ SDK RELEASES
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SUMMARY

Backend engineer and technical lead with 8 years of software experience, including 6+ years building and scaling a high-concurrency transaction platform from 50k to 400k+ daily active users. Promoted from engineer to senior and lead, with hands-on experience in distributed workflows, platform integrations, data pipelines, and production reliability.

EXPERIENCE

SNSoft Sdn Bhd

- Lead Software Engineer (Backend Team Lead)

Dec 2025 – Present

 - Lead a **4-engineer backend team** across six module domains and coordinate service contracts with Frontend, BI, Payment, Provider, and AI teams.
 - Led the rebuild of the core transaction workflow service using **NestJS, gRPC, and Temporal**, processing millions of financial operations daily.
 - Designed **idempotent transaction handling** so retries return the original result without creating duplicate records.
 - Published a self-service integration SDK with **100+ releases**, reducing new request-type onboarding from 30-60 minutes to around 2 minutes.
 - Improved production observability with Grafana, Alibaba Cloud SLS, OpenTelemetry, and Prometheus; optimized queries from one-day timeouts to **30-day analysis windows**.
 - Built an account-change ingestion pipeline processing **10M+ records daily** from 10+ heterogeneous sources using streaming and durable workflows.
- Senior Software Engineer

Jan 2024 – Dec 2025

 - Helped scale the platform as daily active users grew from **50k to 400k+** through Redis Pub/Sub fan-out, atomic ID generation, and cache redesign, deployed without downtime.
 - Built LiveChat, the company's first production service on the **NestJS/gRPC stack**, then mentored three junior engineers to take independent ownership.
 - Delivered the platform reporting engine before a dedicated Business Intelligence team existed, supporting multi-view analytics across **million-row datasets**.
 - Established reusable patterns for configuration, caching, protocol handling, and observability that were adopted by subsequent services.

Software Engineer

Dec 2019 — Jan 2024

- Co-delivered platform-wide **multi-currency (USDT) payment support** across deposits, withdrawals, and transfers.
- Delivered regulatory filings for **million-row datasets** , including automated file splitting under strict submission deadlines.
- Integrated third-party providers and improved live-data reliability through WebSocket heartbeat, reconnection, and failure handling.
- Built operational reporting capabilities that later became the foundation of the platform reporting engine.

Hitachi Ebworx — Mobile Engineer

Apr 2018 — Oct 2019

- Built and delivered a tablet-based Point of Sale application used by **Great Eastern Singapore insurance agents** in their daily sales workflow.
- Developed responsive web and hybrid mobile features across delivery, bug fixing, testing support, and production deployment.
- Owned QA deployments and **automated the daily release process** to improve speed and consistency.

EARLY CAREER

QA & Field-Test Engineer Intern, OPPO Electronics Malaysia HQ - mobile field testing, troubleshooting, test-case design, and product-quality feedback (2017).
IT & Network Support Intern, Chin Hin - IT and network support across 23 branches (2014).

SKILLS

LANGUAGES	TypeScript · JavaScript · SQL · Java (working) · Rust (working)
BACKEND	Node.js · NestJS · gRPC · Protocol Buffers · Express · WebSocket · Temporal · Apache
DATA	MongoDB (aggregation, indexing, Atlas Search) · PostgreSQL · Redis (Pub/Sub, Cluster) · ClickHouse · Hologres · Apache Flink · change data capture
OBSERVABILITY & INFRA	Grafana · Alibaba Cloud SLS · OpenTelemetry · Prometheus · HashiCorp Vault · Docker · GitHub Actions · Alibaba Cloud
QUALITY	Jest · Testcontainers · JUnit 5 · JaCoCo · integration testing · coverage gates · release regression
AI ENGINEERING	Claude Code · MCP · AI-assisted delivery workflows · context engineering · LLM application development
PRACTICES	contract-first APIs · event-driven architecture · idempotent processing · saga & compensation · domain-driven design · zero-downtime migrations

EDUCATION

Bachelor of Information Technology - HELP University · 2017 · English · Mandarin · Malay